1. BASE-MODEL PERFORMANCE OF RECENT OPEN-WEIGHT LLMS

To characterize how far current general-purpose models get on this task without any task-specific training, we evaluated Qwen3.8-27B and Qwen3.5-9B as base models on the PV-Miner test set. These two backbones were newly added in this work to test whether the proposed preference optimization remains effective on recent-generation models.

The evaluation uses the same test split, the same prompt, the same metrics and the same decoding configuration as all other experiments reported in the main text (temperature $= 0$, no sampling, non-thinking mode, max 1024 generated tokens). No generation reached the output length limit, so the low recall reported below is not an artifact of truncation. Results are from a single run.

Table S1. Base-model (no task-specific training) performance on the PV-Miner test set ($n = 248$ messages), in %.

	Code			Sub-code			Span		
Model	P	R	F1	P	R	F1	P	R	F1
Qwen3.8-27B	79.18	62.57	69.90	70.61	50.97	59.20	84.64	41.80	55.96
Qwen3.5-9B	78.83	39.06	52.24	60.91	34.67	44.19	85.07	31.26	45.72

Both models show a pronounced precision–recall asymmetry: precision is moderate to high at all three levels, while recall is low, most severely at the Span level (31.26% and 41.80%). Without exposure to the PV-Miner codebook, the base models produce a small number of annotations that are often locally plausible but leave most of the annotated content unrecovered. Supervised fine-tuning raises Span F1 for the same two backbones to 87.02% and 88.27% (Table 2 in the main text).

Base-model and prompting-based results for Llama-3.3-70B, Llama-3.1-8B, Llama-3.2-3B and Qwen2.5-1.5B were reported in the PVminerLLM study [1], which evaluated prompting and supervised fine-tuning on the same benchmark and the same train/test split.

2. EXTENDED DPO VARIANT DISCUSSION

Table S2. Comparison of PVminerLLM2 preference optimization with existing methods. Prior approaches rely on completion-level signals, which can dilute learning under low edit-distance settings. PVminerLLM2 introduces token-level supervision and confidence-gated stabilization, focusing updates on semantically critical tokens (e.g., labels and spans). ✓: supported; –: not supported.

Method	**Signal**	**Token imp.**	**Low edit dist.**	**Calib./stability**	**Critical tokens**
DPO	Completion	–	–	✓ (ref. reg.)	–
SimPO	Completion	–	–	✓ (implicit reward)	–
IPO / ΨPO	Completion	–	–	✓ (margin regr.)	–
Cal-DPO	Completion	–	–	✓ (calibration)	–
DPO-Positive	Completion	–	Partial	✓ (chosen lik.)	–
PVminerLLM2 (Ours)	**Token + compl.**	✓	✓	✓ **(conf.-gated)**	✓

Preference-based alignment. Modern preference-based alignment improves controllability by learning from pairwise judgments, traditionally via RLHF-style pipelines and reward modeling, and more recently via direct offline objectives that avoid on-policy RL instability and cost [2]. For structured clinical annotation, however, small token-level errors (e.g., a single label token or a Span boundary) can invalidate an otherwise well-formed output, making token-critical supervision central to reliable deployment.

Direct Preference Optimization (DPO). DPO optimizes a policy to prefer chosen completions over rejected ones using an offline, reference-regularized objective [3]. In low-edit-distance regimes,

DPO can be brittle because the learning signal is aggregated at the completion level: near-identical pairs yield weak separation, and uniform sequence-level weighting spreads gradient over shared structural tokens instead of the few decisive semantic tokens. *Our approach differs:* the preference optimization used in PVminerLLM2 explicitly **reallocates learning signal to the few token positions that encode correctness** (e.g., Code/Sub-code/Span and differing tokens) and adds a **confidence-gated stabilization term** to prevent under-confident semantic tokens from drifting.

Reference-free preference objectives (SimPO). SimPO removes the explicit reference model and uses an implicit reward based on the policy's own average token log-likelihood, improving simplicity and efficiency while still encouraging preference separation [4]. In structured extraction, a completion-average reward can still be dominated by high-frequency scaffolding tokens, leaving rare label/Span tokens under-updated. *Our approach differs:* the preference optimization in PVminerLLM2 maintains preference learning while **making the supervision token-selective**, so improvements concentrate on sparse semantic fields rather than being diluted by schema scaffolding.

Margin-targeting reformulations (IPO / ΨPO). ΨPO provides a unifying view of preference objectives, and IPO instantiates a stable regression-style alternative to logistic classification that can help avoid collapsed margins [5]. Yet IPO remains completion-level: it still relies on a single scalar signal per completion, which can be insufficient when only 1–3 tokens determine correctness. *Our approach differs:* the preference optimization in PVminerLLM2 complements margin targeting with **token-adaptive weighting plus a confidence-triggered anchor** that directly increases the absolute likelihood of under-confident label/Span tokens.

Calibration-based regularization (Cal-DPO). Cal-DPO argues that contrastive preference learning can ignore the absolute scale of implicit rewards and introduces a calibration term to mitigate mis-scaling and some forms of likelihood squeezing [6]. However, its regularization is still applied at the completion level, without distinguishing which tokens are semantically decisive. *Our approach differs:* the preference optimization in PVminerLLM2 uses **conditional token-level regularization** that activates only when the policy is under-confident, and can be restricted to **high-value semantic fields**, matching hierarchical labels and Span extraction.

Preventing chosen-likelihood degradation (DPO-Positive). DPO-Positive highlights a pathology where standard DPO can increase the relative preference margin while decreasing the absolute likelihood of the chosen completion, and introduces a corrective penalty to protect chosen likelihood [7]. This is especially relevant under low-edit-distance preferences, but the correction remains completion-level and does not directly address token-importance skew. *Our approach differs:* the preference optimization in PVminerLLM2 provides a **stronger, more localized guarantee** by enforcing a **token-wise floor on under-confident semantic tokens** (labels/Spans), concentrating corrective signal exactly where structured errors occur.

Summary of the core distinction. Across recent DPO-style variants, the optimization signal is primarily driven by a small number of completion-level scalars (margins, logits, or calibrated ratios). The preference optimization framework used to train PVminerLLM2 is distinct in being **token-first**: it (i) concentrates updates on semantically critical tokens via value-aware reweighting, and (ii) stabilizes structured generation with a confidence-gated, SFT-anchored barrier that activates only when needed. This combination is designed to be robust to *low-separation* preferences and *token-importance skew*, which are common in hierarchical label and evidence Span extraction.

3. CODEBOOK

The annotation schema consists of eight major Codes, each representing a distinct communicative or social construct within the patient voice. Each major Code has a corresponding set of Sub-codes that capture more specific communicative intents. Below are concise summaries of these key categories:

1. Social Determinants of Health (SDOH). Refers to the process of sharing and seeking knowledge about the social, economic, and environmental factors that significantly influence an individual's health and well-being. These include economic stability, access and quality to education, access and quality of health care, neighborhood and built environment, and social and community context. *Sub-codes:*

- **Economic Stability:** Financial security and resources to afford healthcare, housing, food, and other necessities.

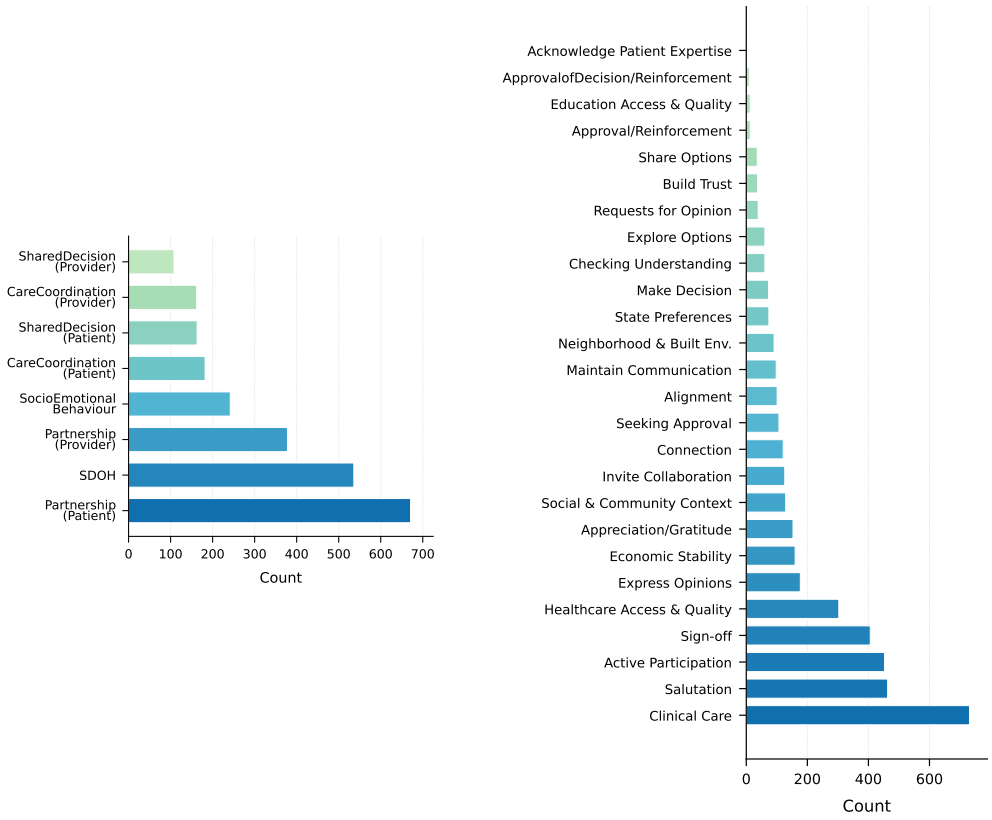


Fig. S1. Distribution of Codes (left) and Sub-codes (right). The image is in high resolution, please zoom in for better vision. *Alt-text: Two horizontal bar charts of annotation frequency. The left chart shows the counts of the eight Codes and the right chart shows the counts of the Sub-codes, each sorted from most to least frequent. Both charts are long-tailed, with a small number of high-frequency categories and many low-frequency ones.*

- **Education Access and Quality:** Provide information on the availability and effectiveness of educational opportunities, the quality of education, including resources, teaching standards, and curriculum.
- **Access and quality to healthcare:** The ability to obtain the necessary health services and the effectiveness and standard of care that patients receive.
- **Neighborhood and Built Environment:** Physical and social surroundings such as housing, transportation, safety, and environmental quality.
- **Social and Community Context:** Social and community context refers to relationships, interactions, and conditions within the environments where people live, work, and interact, and it significantly influences health outcomes.

2. Partnership from the Patient Side. Patient partnership involves establishing and strengthening the alliance between patients and healthcare providers through active participation, open communication, and mutual respect. *Sub-codes:*

- **Active Participation/Involvement:** Patient is active in providing information to aid diagnosis and problem solving, priorities for treatment or management, asking questions and/or contributing to the identification of management approaches.
- **Express Opinions:** Patients actively share their thoughts, concerns about their care and treatment and provide feedback on their experiences with healthcare services.
- **Signoff:** Courteous closure that marks the completion of a message.

- **State Preferences:** Individual values, desires, and priorities of the patient regarding their healthcare.
- **Alignment:** Refers to establishment of a meaningful, trust-based relationship between the patient and healthcare provider.
- **Appreciation/Gratitude:** appreciation and gratitude that are expressed when patients acknowledge the care and support they receive.
- **Connection:** Information that is not directly related to the medical issue being discussed, strengthening the relationship between the patient and provider.
- **Salutation:** Greeting or addressing the provider by name or title.
- **Clinical Care:** Refers to the patient's experiences, perceptions, and self-reported expressions concerning symptoms, diagnoses, treatments, or medical procedures they receive or seek.
- **Build Trust:** Fostering a sense of confidence and reliability, wherein the patient feels assured that the provider is acting in their best interest.

3. Partnership from the Provider Side. Refers to fostering a collaborative and equitable relationship between healthcare providers and patients, involving the equalization of status and ensuring that patients feel valued and empowered. *Sub-codes:*

- **Invite Collaboration:** Inviting patients to participate in decisions related to their condition through informed consent, treatment planning, and self-management.
- **Requests for Opinion:** actively seeking patients' perspectives and preferences on treatment options, care plans, and health-related decisions.
- **Checking Understanding/Clarification:** Confirming that the patient fully understands the information being communicated, including key concepts related to their condition, treatment options, costs, and care plan.
- **Appreciation/Gratitude:** Refers to the expression of appreciation and gratitude by healthcare providers when acknowledging patients' engagement, cooperation, and participation in their care.
- **Signoff:** Courteous ending signaling completion of communication.
- **Acknowledge Patient Expertise/Knowledge:** Recognizing and valuing the insights patients gain from their lived experiences with their health conditions.
- **Maintain Communication:** It involves keeping patients informed and engaged throughout the care process by clearly communicating that additional information or updates will be provided at a later time, ensuring patients are aware of what to expect.
- **Alignment:** Confirming or seeking confirmation that the patient and provider share a mutual understanding and perspective on a given issue.
- **Connection:** Comments or information unrelated to the medical issue that serve to strengthen the patient–provider relationship and foster rapport.
- **Salutation:** Refers to the act of greeting or respectfully addressing the patient.
- **Clinical Care:** Refers to the provider's planning, coordination, and delivery of diagnoses, treatments, and medical procedures tailored to the patient's clinical needs.
- **Build Trust:** Involves fostering a relationship with the patient characterized by mutual respect, transparency, and reliable communication, which is essential for effective healthcare delivery.

4. Shared Decision-Making from the Patient Side. Involves patients actively engaging in treatment discussions, asking questions, and providing informed approval of decisions that align with their values and goals, while fostering confidence in the chosen course of action. *Sub-codes:*

- **Explore Options:** Indicators of a patient's interest in learning about all options include asking detailed questions, expressing curiosity about alternatives, seeking clarification, discussing how options align with personal goals, and requesting additional resources for further information.
- **Seeking Approval:** Refers to the patient asking for or seeking permission from the health-care provider.
- **Approval of Decision:** Affirmation refers to the patient agreeing with a chosen plan or decision after understanding the available options and their implications.

5. Shared Decision-Making from the Provider Side. Refers to statements or questions that create an opportunity for a two-way exchange of information regarding treatment options, evaluation of decisions, or consideration of alternatives. *Sub-codes:*

- **Share Options:** Refers to the process of identifying, presenting, and deliberating the available options for a specific decision or problem, encompassing a discussion of their associated risks, benefits, and potential outcomes.
- **Make Decision:** Refers to the collaborative process in which the provider and patient jointly select a course of action after considering all available options, associated risks and benefits, and the patient's personal preferences.
- **Approval of Decision/Reinforcement:** Refers to reinforcing a decision by providing positive feedback or validating the patient's involvement in the decision-making process.

6. Socioemotional Behavior. Refers to responding to patients' emotional cues by validating their feelings, offering empathy, reassurance, and encouragement, acknowledging errors transparently, and providing positive reinforcement, reward, or approval. No Sub-codes are defined.
7. Care Coordination Provider. Refers to the deliberate organization and integration of healthcare services by the provider to ensure that patient care is delivered effectively, efficiently, and in a timely manner. No Sub-codes are defined.
8. Care Coordination Patient. Represents the patient's active engagement in communicating and coordinating care among various healthcare providers and services to ensure continuity and effectiveness. No Sub-codes are defined.

These hierarchical categories enable PVminer to capture both the content and the relational intent embedded in patient discourse, supporting downstream analyses that link language use to social context, decision-making, and health outcomes.

4. EXAMPLES FROM ALL FOUR DATA SOURCES

Table S3 presents examples from each of the four data sources.

Table S3. Examples of patient voice extraction from the PV-Miner dataset. Each annotation is a (Code, Sub-code, Span) triple shown in JSON-style format. Sources include Yale New Haven Hospital (YNHH) secure messages, Woven Health messages, Bethesda interviews, and survey responses. "None" means the Code has no Sub-codes.

Source	Message (Input)	Extracted Triples (Output)
YNHH	Do I call them or do they reach out when they get the referral?	`[{"code": "PartnershipPatient", "subcode": "activeParticipation", "span": "Do I call them"}, {"code": "SharedDecisionPatient", "subcode": "ExploreOptions", "span": "or do they reach out"}, {"code": "CareCoordinationPatient", "subcode": "None", "span": "they get the referral"}]`
Woven	Needs refill on montelukast 10 mg tablet lisinopril 5 mg tablet	`[{"code": "PartnershipPatient", "subcode": "activeParticipation", "span": "Needs refill on"}, {"code": "PartnershipPatient", "subcode": "ClinicalCare", "span": "montelukast 10 mg tablet"}, {"code": "PartnershipPatient", "subcode": "ClinicalCare", "span": "lisinopril 5 mg tablet"}]`
Bethesda	I've paid for my labs and will take care of it next week. Would it be possible to have Diflucan refilled now. I have a yeast infection.	`[{"code": "SDOH", "subcode": "EconomicStability", "span": "I've paid for my labs"}, {"code": "PartnershipPatient", "subcode": "activeParticipation", "span": "will take care of it"}, {"code": "SharedDecisionPatient", "subcode": "SeekingApproval", "span": "Would it be possible to have"}, {"code": "PartnershipPatient", "subcode": "ClinicalCare", "span": "Diflucan refilled now"}, {"code": "PartnershipPatient", "subcode": "ClinicalCare", "span": "I have a yeast infection"}]`
Survey	Should have been advised sooner.	`[{"code": "PartnershipPatient", "subcode": "expressOpinions", "span": "Should have been advised"}]`

5. PREFERENCE PAIR CONSTRUCTION DETAILS

Algorithm S1 details the confusion-aware preference pair construction procedure described in Section 3.4 of the main text. The confusion matrix $\mathbf{M}$ is built by running the SFT model on a held-out validation set and recording, for each gold label, which incorrect labels the model predicts most frequently. For each gold-annotated example, we generate three dispreferred outputs by applying all three perturbation types: label confusion, deletion, and insertion. Label confusion replaces a Code or Sub-code with a confused alternative drawn from $\mathbf{M}$, weighted by empirical confusion frequency; deletion removes a randomly selected tuple; and insertion adds a spurious but schema-valid tuple whose Code is sampled proportionally to confusion frequency. In total, this yields up to $3|\mathcal{D}_{\text{train}}|$ pairs (fewer when single-tuple examples skip deletion). All dispreferred outputs preserve JSON schema validity.

6. FULL SUB-CODE PERFORMANCE TABLE

Table S4 presents Sub-code level performance comparison between PVminerLLM F1 and PVminer-LLM2 F1. ΔF1 denotes improvement. It is sorted by the gain. Three Sub-codes: Approval/Reinforcement (13), ApprovalofDecision/Reinforcement (10), and acknowledgePatientExpertiseKnowledge (1) remain at zero F1 under both PVminerLLM and PVminerLLM2, as their extremely low instance counts fall below the minimum frequency needed for either SFT or preference optimization to learn meaningful representations. Annotation of the dataset is ongoing, and future releases will include additional instances for these underrepresented categories.

7. PROMPT FOR TRAINING THE LLM

Because either SFT or alignment is training based on instruction, we provide our designed prompt for this task here.

Table S4. Sub-code level performance comparison between PVminerLLM and PVminerLLM2. ΔF1 denotes improvement. It is sorted by the ΔF1.

Class	Freq	PVminerLLM F1	PVminerLLM2 F1	ΔF1
checkingUnder standing/clarification	61	11.11	**38.71**	+27.60
Neighborhood AndBuiltEnvironment	91	28.57	**48.65**	+20.08
EducationAccess AndQuality	13	66.67	**85.71**	+19.04
requestsForOpinion	39	28.57	**47.06**	+18.49
ExploreOptions	61	14.29	**31.58**	+17.29
SeekingApproval	107	33.33	**50.00**	+16.67
connection	121	65.12	**77.55**	+12.43
SocialAnd CommunityContext	129	70.59	**81.36**	+10.77
expressOpinions	177	58.67	**69.14**	+10.47
activeParticipation /involvement	453	69.41	**78.69**	+9.28
HealthCareAccess AndQuality	303	63.64	**72.88**	+9.24
inviteCollabration	126	53.06	**60.61**	+7.55
ShareOptions	36	22.22	**28.57**	+6.35
MakeDecision	73	40.00	**44.44**	+4.44
alignment	101	91.67	**96.00**	+4.33
maintainCommunication	98	60.87	**63.41**	+2.54
salutation	463	98.97	**100.00**	+1.03
Appreciation/Gratitude	153	79.31	**79.41**	+0.10
Clinical Care	731	77.17	**77.27**	+0.10
Approval/Reinforcement	13	0.00	0.00	+0.00
ApprovalofDecision /Reinforcement	10	0.00	0.00	+0.00
acknowledgePatient ExpertiseKnowledge	1	0.00	0.00	+0.00
signoff	406	**92.74**	91.02	-1.72
statePreferences	74	**60.87**	56.00	-4.87
EconomicStability	160	**83.33**	77.33	-6.00
build trust	37	**50.00**	33.33	-16.67

Algorithm S1. Confusion-Aware Preference Pair Construction

Require: Training set $\mathcal{D}_{\text{train}}$ with gold annotations $\{(x_i, Y_i^+)\}$, SFT model π_{sft}, validation set $\mathcal{D}_{\text{val}}$, valid schema $\mathcal{S}$
Ensure: Preference pair set $\mathcal{P}$

// Step 1: Build confusion matrix from SFT errors
Initialize confusion matrix $\mathbf{M}[c_{\text{gold}}, c_{\text{pred}}] \leftarrow 0$ for all Code/Sub-code labels
for each $(x, Y^+) \in \mathcal{D}_{\text{val}}$ **do**
  $\hat{Y} \leftarrow \pi_{\text{sft}}(x)$
  for each gold tuple $(c_g, s_g, t_g) \in Y^+$ **do**
    if matched prediction $(\hat{c}, \hat{s}, \hat{t})$ exists with $\hat{c} \neq c_g$ or $\hat{s} \neq s_g$ **then**
      $\mathbf{M}[c_g, \hat{c}] \mathrel{+}= 1; \quad \mathbf{M}[s_g, \hat{s}] \mathrel{+}= 1$
Normalize rows of $\mathbf{M}$ to obtain confusion distributions

// Step 2: Generate preference pairs
$\mathcal{P} \leftarrow \emptyset$
for each $(x_i, Y_i^+) \in \mathcal{D}_{\text{train}}$ **do**
  // Perturbation 1: Label confusion
  Select tuple j uniformly from Y_i^+
  Sample replacement label $\tilde{c}_j \sim \mathbf{M}[c_j, \cdot]$ (or $\tilde{s}_j \sim \mathbf{M}[s_j, \cdot]$)
  $Y_{i,1}^- \leftarrow Y_i^+$ with (c_j, s_j, t_j) replaced by $(\tilde{c}_j, \tilde{s}_j, t_j)$
  $\mathcal{P} \leftarrow \mathcal{P} \cup \{(x_i, Y_i^+, Y_{i,1}^-)\}$

  // Perturbation 2: Deletion
  if $|Y_i^+| \geq 2$ **then**
    Select tuple k uniformly from Y_i^+
    $Y_{i,2}^- \leftarrow Y_i^+ \setminus \{(c_k, s_k, t_k)\}$
    $\mathcal{P} \leftarrow \mathcal{P} \cup \{(x_i, Y_i^+, Y_{i,2}^-)\}$

  // Perturbation 3: Insertion
  Sample spurious Code $\tilde{c} \sim \mathbf{M}[\cdot]$ weighted by confusion frequency
  Sample valid Sub-code $\tilde{s}$ for $\tilde{c}$ from $\mathcal{S}$
  Sample span $\tilde{t}$ from a random contiguous phrase in x_i
  $Y_{i,3}^- \leftarrow Y_i^+ \cup \{(\tilde{c}, \tilde{s}, \tilde{t})\}$
  $\mathcal{P} \leftarrow \mathcal{P} \cup \{(x_i, Y_i^+, Y_{i,3}^-)\}$

  Verify all $Y_{i,\cdot}^-$ are schema-valid under $\mathcal{S}$; serialize as JSON
return $\mathcal{P}$

Prompt - Engineered Prompt

- XML Structuring
- Chain-of-Thought (4-step reasoning)
- Self-Validation (quality gate)
- Decision Logic (direction-aware)
- Disambiguation Rules
- Performance Targets

```
<role>
Expert patient-centered communication analyst with >95% accuracy in medical message
multi-label classification.
</role>

<performance_target>
CRITICAL REQUIREMENTS:
- Code Accuracy: >95%
- Sub-code Accuracy: >95%
- Span Accuracy: >98% (character-perfect)
```

```
Every annotation must be defensible and verification-validated.
</performance_target>

<task>
Extract Code, Sub-code, and Span triples from medical secure messages.

INPUT:
- Context (the message text)
- Message Direction (TO_PAT_YN)

OUTPUT:
- JSON list of {Code, Sub-code, Span} objects

CONSTRAINT:
- MULTI-LABEL task (one message may contain multiple valid triples)
</task>

<message_direction>
CRITICAL: Message direction determines Code selection.

- TO_PAT_YN = "Y": Provider speaking TO patient
- TO_PAT_YN = "N": Patient speaking TO provider

USE CASES:
- Provider to patient: Use PartnershipProvider, SharedDecisionProvider,
  CareCoordinationProvider when applicable
- Patient to provider: Use PartnershipPatient, SharedDecisionPatient,
  CareCoordinationPatient when applicable
- SDOH and SocioEmotionalBehaviour apply regardless of direction
</message_direction>

<critical_rules>
RULE VIOLATIONS RESULT IN ANNOTATION FAILURE

1. Span Source:
   - Extract Spans ONLY from the provided message text
   - Context is for understanding only
   - Never invent, paraphrase, or infer Spans

2. Code and Sub-code Validity:
   - Every Sub-code MUST be valid for its Code
   - If a pairing is illogical or invalid, loop back and re-select

3. Span Exactness:
   - Copy EXACT text from the message
   - Preserve punctuation, capitalization, and spacing
   - No paraphrasing

4. Multi-label Requirement:
   - Identify ALL relevant Code and Sub-code pairs in the message
</critical_rules>

<reasoning_process>
Follow this 4-step verification process:

STEP 1: CONTEXT AND DIRECTION ANALYSIS
- Read the full message carefully
- Determine message direction using TO_PAT_YN
- Understand speaker intent and conversational goal

STEP 2: PHRASE DECOMPOSITION AND CODE MATCHING
- Break the message into semantic units (phrases or clauses)
- For each phrase, identify intent:
  * SDOH
  * PartnershipProvider or PartnershipPatient
  * SharedDecisionProvider or SharedDecisionPatient
  * CareCoordinationProvider or CareCoordinationPatient
  * SocioEmotionalBehaviour
- Use TO_PAT_YN to select Provider vs Patient variants
- Match each phrase to the correct Code definition
- Verify Code and Sub-code pairing is logical and valid

STEP 3: SPAN EXTRACTION AND VERIFICATION
- Extract the MINIMUM complete supporting phrase
- Spans must come ONLY from the message text
- Verify character-level exactness
- If the Span does not exist exactly, reject the annotation

STEP 4: CROSS-VALIDATION (MOST IMPORTANT)
```

```
Verification priority:
1. Best semantic match confirmed (if not, loop back to Step 2)
2. Sub-code valid for Code
3. Span is exact and present in message
4. All relevant phrases analyzed
5. Disambiguation rules applied correctly
6. High-confidence annotation defensible to experts
</reasoning_process>

<codes_definitions>
The following are authoritative ground-truth definitions. Names must match exactly.
(Full list omitted here for brevity).

FORMAT (Code WITH Sub-codes):
CODE_NAME: <one-sentence operational definition>.
|- SUBCODE_1: <one-sentence operational definition>.
|- SUBCODE_2: <one-sentence operational definition>.
|- SUBCODE_K: <one-sentence operational definition>.

FORMAT (Code WITHOUT Sub-codes):
CODE_NAME: <one-sentence operational definition>.
|- None: No sub-codes are defined for this Code.
</codes_definitions>

<disambiguation_rules>
Apply systematically to resolve ambiguity.

Salutation vs Signoff:
- Opening greetings indicate salutation
- Closing phrases indicate signoff
- Position determines classification

Appreciation/Gratitude vs Signoff:
- Simple closing thanks indicates signoff
- Specific appreciation indicates Appreciation/Gratitude

Provider vs Patient Codes:
- Use TO_PAT_YN strictly
- TO_PAT_YN = "Y" -> Provider codes
- TO_PAT_YN = "N" -> Patient codes

SharedDecision Codes:
- Use TO_PAT_YN to select Provider vs Patient variants

SDOH Sub-code Selection:
- EconomicStability: finances, income, food, housing
- EducationAccessAndQuality: education, literacy
- HealthCareAccessAndQuality: access to care, insurance, physical activity
- NeighborhoodAndBuiltEnvironment: housing, transportation, environment
- SocialAndCommunityContext: social support, isolation, discrimination

CareCoordination vs maintainCommunication:
- maintainCommunication: future updates only
- CareCoordination: concrete coordination with other providers

requestsForOpinion vs inviteCollaboration:
- requestsForOpinion: asks patient views
- inviteCollaboration: invites joint participation

SocioEmotionalBehaviour:
- Emotional support, reassurance, empathy, politeness
- Only Sub-code "None" is valid
</disambiguation_rules>

<output_format>
Return JSON with a "results" array:

{
  "results": [
    {
      "Code": "exact Code name",
      "Sub-code": "exact Sub-code name",
      "Span": "EXACT text from message"
    }
  ]
}

If no annotations apply:
{"results": []}
```

```
</output_format>

<quality_gate>
MANDATORY verification before submission:

1. JSON is parseable
2. All Sub-codes valid for Codes
3. All Spans are exact and present in message
4. Best semantic match verified
5. All disambiguation rules applied
6. High confidence suitable for expert review

Accuracy is paramount. Quality over speed.
</quality_gate>

INPUT:
TO_PAT_YN: N (Patient speaking to provider)

Context:
Dr. Person1 I need my prescription sent to the pharmacy for my flecainide acetate
100 mg tablets twice a day the pharmacist has try requesting it no success and I
don't have any pills. Person2
```

REFERENCES

1. S. Fodeh, L. Ma, G. Puthiaraju, *et al.*, "Pvminerllm: Structured extraction of patient voice from patient-generated text using large language models," arXiv preprint arXiv:2603.05776 (2026).
2. L. Ouyang, J. Wu, X. Jiang, *et al.*, "Training language models to follow instructions with human feedback," arXiv preprint arXiv:2203.02155 (2022).
3. R. Rafailov, A. Sharma, E. Mitchell, *et al.*, "Direct preference optimization: Your language model is secretly a reward model," Adv. Neural Inf. Process. Syst. **36** (2023).
4. Y. Meng, M. Xia, and D. Chen, "Simpo: Simple preference optimization with a reference-free reward," (2024).
5. M. G. Azar, Z. D. Guo, B. Piot, *et al.*, "A general theoretical paradigm to understand learning from human preferences," in *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, vol. 238 (PMLR, 2024), pp. 4447–4455.
6. T. Xiao, Y. Yuan, H. Zhu, *et al.*, "Cal-dpo: Calibrated direct preference optimization for language model alignment," (2024). Accepted by NeurIPS 2024 Main.
7. A. Pal, D. Karkhanis, S. Dooley, *et al.*, "Smaug: Fixing failure modes of preference optimisation with dpo-positive," (2024).